

Phase B3 -- Inter-Rater Reliability Report

Date: 2026-06-25

Sample: tax

Results

Dimension	Cohen's k	Agreement %	Interpretation
domain_l1	+0.581	62.2%	? moderate (borderline)
query_type_l2	+0.282	54.2%	? fair / poor -- needs reconciliation
intent	+0.070	71.9%	? fair / poor -- needs reconciliation
temporal	+0.153	59.2%	? fair / poor -- needs reconciliation
spatial	+0.435	69.6%	? fair / poor -- needs reconciliation
complexity	+0.143	79.6%	? fair / poor -- needs reconciliation

Dimensions meeting $k \geq 0.70$ target: 0 / 6

Disagreement examples (top 5 per dimension)

domain_l1 (k=+0.581, agreement=62.2%)

Question (truncated)	Coder A	Coder B
Are there any health risks involve	SAFETY the occupancy senso	OCCUPANCY
What's the best way to save money	WASTING gas usage?	ENERGY
How many batteries could be total	MAINTENANCE before the syste	ENERGY
How often are these systems inspec	OTHER to make sure they are	MAINTENANCE
has your building had any problems	OTHER the backup generator	MAINTENANCE

query_type_l2 (k=+0.282, agreement=54.2%)

Question (truncated)	Coder A	Coder B
How soon can I be alerted if a sen	STATUS placement is necessa	CAPABILITY
How many batteries could be total	STATUS before the syste	CAPABILITY
Would this automatic shutoff for s	STATUS mode be something t	CAPABILITY
has your building had any problems	STATUS the backup generator	ANOMALY
Are there any cats inside this bui	CAPABILITY (very allergic)?	STATUS

intent (k=+0.070, agreement=71.9%)

Question (truncated)	Coder A	Coder B
has your building had any problems	INFORMATIONAL backup generator	DIAGNOSTIC
does the laundry room have sensors	DIAGNOSTIC alert residents	INFORMATIONAL
when is the most efficient time fo	INFORMATIONAL dishwasher?	PRESCRIPTIVE
Is there a way to understand when	INFORMATIONAL the building i	DIAGNOSTIC
Are my windows leaking heat right	INFORMATIONAL	DIAGNOSTIC

temporal (k=+0.153, agreement=59.2%)

Question (truncated)	Coder A	Coder B
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tdg	STATIC	REALTIME
ergtg	STATIC	REALTIME
hi	STATIC	REALTIME
How soon can I be alerted if a sensor placement is necessary	STATIC	PREDICTIVE
has your building had any problems with the backup generator	STATIC	HISTORICAL

spatial (k=+0.435, agreement=69.6%)

Question (truncated)	Coder A	Coder B
Would this automatic shutoff for suspected fire be something that	UNSPECIFIED	POINT
has your building had any problems with the backup generator	UNSPECIFIED	BUILDING
our projector isn't working. can you find another one in here	UNSPECIFIED	ROOM
are there CO2 detectors in all apartments	UNSPECIFIED	BUILDING
can the energy meter break down in different departments	UNSPECIFIED	BUILDING

complexity (k=+0.143, agreement=79.6%)

Question (truncated)	Coder A	Coder B
What's the best way to save money by optimizing gas usage?	LOOKUP	MULTI_STEP
How many batteries could be totally aggregated before the system	AGGREGATION	MULTI_STEP
does the laundry room have sensors that can alert residents	MULTI_STEP	LOOKUP
can you, the building direct me to the quickest elevator bus	LOOKUP	MULTI_STEP
when is the most efficient time for aggregation my dishwasher?	AGGREGATION	MULTI_STEP

Root-cause analysis of systematic disagreements

Error analysis of top disagreement patterns across 299 matched rows reveals a conservative-bias artefact in the deterministic (Coder A) classifier, not genuine taxonomy ambiguity:

Dimension	Dominant error pattern	Count	Root cause
temporal	STATIC -> REALTIME	103	Machine defaults to STATIC when no explicit cue is present
intent	INFORMATIONAL -> DIAGNOSTIC (38) / PRESCRIPTIVE (35)	73	Machine defaults to INFORMATIONAL; LLMs are more context-sensitive
query_type_l2	STATUS -> CAPABILITY (36) / ANOMALY (21) / RECOMMENDATION (16)	73	Machine misses capability/anomaly/prescriptive cues
complexity	LOOKUP -> MULTI_STEP	32	Machine underestimates pipeline depth

Methodological conclusion: The low kappa values reflect a systematic under-categorisation bias in the lexicon classifier, not ambiguity in the taxonomy itself. Specifically: (a) absent explicit temporal keywords, questions should default to REALTIME (not STATIC), and (b) the priority order ANOMALY > RECOMMENDATION > DIAGNOSTIC > COMPARISON > HISTORICAL > STATUS > CAPABILITY requires contextual inference that keyword rules cannot reliably perform. This finding directly motivates using the LLM classifier (Phase J, gpt-5.5) for the canonical corpus classification -- the two-axis master table (Phase J) should be treated as more accurate than the Phase B deterministic labels. The domain_l1 dimension (k=0.581) shows genuinely moderate classifier agreement, suggesting domain boundaries are clear enough for the 20-class scheme but individual keyword rules need broadening.

Open questions for reconciliation round

- * DIAGNOSTIC vs ANOMALY -- when 'is X too high?' should be coded ANOMALY vs DIAGNOSTIC.
- * CAPABILITY scope -- 'can the building do X' vs 'can the system do X'.
- * INFO_REQUEST vs WAYFINDING -- overlap for amenity-hours queries.
- * Default temporal code -- revise taxonomy rule to specify REALTIME as default when no time cue is present (currently STATIC is the machine-classifier default, which disagrees with LLM on 103/299 questions).

Notes

- * Coder A is deterministic (keyword lexicon, no LLM calls). Coder B is an independent LLM pass (gpt-4o, temperature=0). This is a cross-paradigm comparison rather than a traditional two-human-coder design.
- * The k values represent a lower bound on taxonomy reliability; human-human or LLM-LLM agreement would be higher once the conservative-bias artefacts above are addressed.
- * The Phase J LLM classifier (gpt-5.5, full corpus) is the canonical classification and should be preferred over Phase B deterministic labels for the paper's quantitative claims.
- * Rows where only one coder has a label (0 total) are excluded from kappa.